CHAPTER THREE
LANGUAGE: MEANING AND DEFINITION

What can go wrong with the use of language in the course of argument?

3.1. THE EMOTIVE AND COGNITIVE MEANING OF WORDS

- Argument is an expression.
- Reasoning is represented in words.
- Logic works in language.
- This chapter focuses on two functions of language. These are:
  - Emotive meaning
  - Cognitive meaning

- Words that are said to have **cognitive meaning** impart/convey information.
- Words that are said to have **emotive meaning** evoke emotions.



Example 1

$7\times 5^{(3x+3)} = 35$

$5^{(3x+3)} = 5$

$\log 5^{(3x+3)} = \log 5$

$3x + 3 = 1$

$3x = -2$

$X = -2/3$

Example 2

The death penalty, which is legal in 36 states, has been carried out most often in Georgia; however, since 1977 Texas holds the record for the greatest number of executions.

Example 3

The death penalty is a cruel and inhuman form of punishment in which hapless prisoners are dragged from their cells and summarily slaughtered only to satiate the bloodlust of a vengeful public.

- **Value Claims** are the cognitive meaning of an emotive statement.
- **Factual Disputes** are disputes on facts
- **Verbal Disputes** are disputes that center on the meaning of a word.

Example1

A. We are not free because our behavior is determined by our genes in conjunction with environmental influences.

B. *I disagree. This is a free country. Ethiopians are a free people!*

Example2

A. The battle of Adwa took place in 1889.

B. No it took place in 1896.

3.2. THE INTENSION AND EXTENSION OF TERMS

- arguments are comprised of statements
- statements are made up of words, words symbolize meanings and meanings are conveyed through definitions.
- the basic units of any ordinary language are words.
- logic is more interested in particular words called **terms.**
- A term is any word or arrangement of words that may serve as the subject of a statement.
- Term includes proper names, common names, and Descriptive phrases.

• Verbs, non-substantive adjectives, adverbs, prepositions, conjunctions and all non-syntactic arrangement of words are NOT terms.

•In identifying a word as a term it is important to distinguish the ***use*** of words from the ***mention*** of a word.

•A word is said to be *used* when it is in its proper function in English grammar.

•Any word can be a term if they are *mentioned*

Example

1. ***"Did"*** *is a three-letter English word. [mentioned]*
2. *I know what you* ***did*** *last night. [used]*

- Words are usually considered to be symbols and the entities they symbolize are usually called meanings.
- Terms, being made up of words, are also symbols, but the meanings they symbolize are of two kinds: **intensional** and **extensional**.
- The **intensional meaning** consists of the qualities or attributes that the term connotes.
- the **extensional meaning** consists of the members of the class that the term denotes.

- The intensional meaning is otherwise known as the **intension** or **connotation**, and the extensional meaning is known as the **extension** or **denotation**.

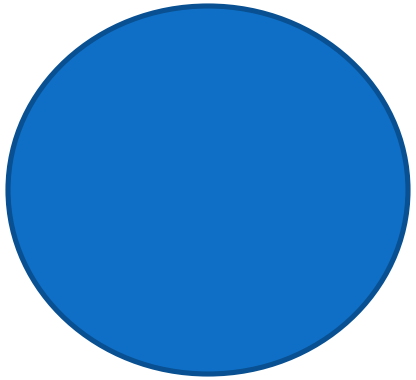
Intension

- Companion animal
- Mammal/domestic
- Four-legged
- Feeds/feasts on rats
- Carnivorous
- Meowing- cry of cats
- Cute
- selfish

Cat

Extension

- All cats in the universe



Examples:

1. ‘dog’ means like Lulu, Bobby, Royal, Cooper, Pluto.

2. Tortoise is like Jonathan, Govinda.

3. Jonathan is the oldest tortoise on earth with the world’s Guinness record of 190 years.

- **Conventional connotation** of a term consists of the properties or attributes that the term commonly connotes to the members of the community who speak the language in question
- If a term denotes an empty (or null) class, the class that has no members then it is said to have **empty extension.**
- The fact that some terms have empty extension indicates that *intension determines extension*. But extension suggests intension.

- **Increasing intension** when each term in the series (except the first) connotes more attributes than the one preceding it
- **Decreasing intension** is the reverse of that of increasing intension.
- **Increasing extension** when each term in the series (except the first) denotes a class having more members than the class denoted by the term preceding it.
- **Decreasing extension** is the reverse of this order.

Examples

1. Tiger; mammal; feline; animal
2. Doctor of medicine; person; brain surgeon; professional person; surgeon
3. Parallelogram; polygon; square; rectangle; quadrilateral
4. Italian sports car; car; vehicle; Maserati; sports car

3.3. DEFINITIONS AND THEIR PURPOSES

"If you wish to converse with me define your terms."

Voltaire

- definition is a group of words that assigns a meaning to some word or group of words.
- every definition consists of two parts: ***definiendum*** and ***definiens***.
- *definiendum* is the word or group of words that is supposed to be defined and the *definiens* is the word or group of words that does the defining.
- There are five types of definitions. These are stipulative, lexical, précising, theoretical, persuasive.

STIPULATIVE DEFINITIONS

- It arbitrarily assigns meaning to a word for the first time. This may involve either coining(make up) a new word or giving a new meaning to an old word.
- It is used to replace a more complex expression with a simpler one.
- It is used to set up secret codes.
- It is often occasioned by some new phenomenon or development.
- It is used as the unique use of language among a particular group of people.
- It cannot be evaluated as true or false.

LEXICAL DEFINITIONS

- is used to report/repeat the meaning that a word already has in a language.
- a lexical definition may be true or false depending on whether it does or does not report the way a word is actually used.
- lexical definitions have the further purpose of eliminating the ambiguity. A word is **ambiguous** when it can be interpreted as having two or more clearly distinct meanings in a given context. ***Examples:*** *party; light; bank; sound; race; right; bachelor, pool*

PRECISING DEFINITIONS

- It's used to reduce the vagueness of a word.
- A word is **vague** if there are borderline cases such that it is impossible to tell whether the word applies to them or not.
- ***Examples:*** *love; happiness; peace; excessive; fresh; rich; poor; normal; discipline vs. abuse;*

THEORETICAL DEFINITIONS

- provides a theoretical picture or characterization of the entity or entities denoted by the *definiendum*.
- it provides a way of viewing or conceiving these entities that suggests deductive consequences.
- Theoretical definition functions as proposals to see or interpret some phenomenon in a certain way.
- Theoretical definitions are neither true nor false.

PERSUASIVE DEFINITIONS

- engender a favorable or unfavorable attitude toward what is denoted by the *definiendum*.
- assigns an emotionally charged or value-laden meaning to a word.
- The objective of a persuasive definition is to influence the attitudes of the reader or listener.
- They are instruments of persuasion.
- ***Example***
- *'Abortion' means the ruthless murdering of innocent human being.*
- *'Abortion' means a safe and established surgical procedure whereby a woman is relieved of unwanted burden.*

3.4.DEFINITIONAL TECHNIQUES

EXTENSIONAL/DENOTATIVE

- Demonstrative (Ostensive) definitions
  - Pointing
- Enumerative definitions
  - Naming them individually
- Definition by Subclass
  - Naming them in group

INTENSIONAL/CONOTATIVE

- Synonymous definition
- Etymological definition
- Operational definition
- Definition by genus and difference

Demonstrative (Ostensive) definitions

- Are probably the most primitive form of definition.
- All one need know to understand such a definition is the meaning of pointing.
- Used if the speakers do not know each other's language.
- Since they are the most primitive they are also the most limited(for instance when the object to be pointed is not available).
- In such definitions the *definiens* is gesture. In sign language the 'words' are signs or gestures.
- They are either partial or complete.

Enumerative definitions

- It assigns a meaning to a term by naming the members of the class the term denotes.
- It names the members individually. The members are single irreducible unit of a larger class.
- They are either partial or complete.

'Planet' means one of the following Mercury, Venus, Earth, Mars, Saturn, Jupiter, Neptune, Uranus, and Pluto.

'Actress' means ……. (finite)

'Number' means …… (infinite)

'Continent' means Africa, Europe, Asia, Australia, North America, and South America. (finite)

'Ocean' means Atlantic, Pacific, Indian, and Antarctic (finite)

Definition by Subclass

- It assigns a meaning to a term by naming subclasses of the class denoted by the term.
- It is used when we encounter something that is either difficult or impossible to completely enumerate.
- It names the members in groups.
- They are either partial or complete.

Example

- *'Fictional work' means either a poem, a play, a novel, or a short story.*

Synonymous

- A **synonymous** definition is one in which the *definiens* is a single word that connotes the same attributes as the *definiendum*. In other words, the *definiens* is a synonym of the word being defined. It is highly concise.

Example: *Physician means doctor.*

Etymological definition

- It's the history of a linguistic form such as a word shown by tracing its development since its earliest recorded occurrence in the language where it is found, by tracing its transmission from one language to another, by analyzing it into its component parts, by identifying its cognates in other languages, or by tracing it and its cognates to a common ancestral form in an ancestral language.

Examples

- 'License' – Latin – *licere* (to be permitted)
- 'Captain' – Latin – *caput* (head)
- 'Principle' – Latin – *principium* (beginning/source)

Operational definition

- An **operational** definition assigns a meaning to a word by specifying certain experimental procedures that determine whether or not the word applies to a certain thing.
- Operational definitions were invented for the purpose of tying down relatively abstract concepts to the solid ground of empirical reality. However, it has the deficiency of conveying only part of the intensional meaning of a term.

Example: *One substance is "harder than" another if and only if one scratches the other when the two are rubbed together.*

Definition by Genus and Difference

- A **Definition by Genus and Difference** assigns a meaning to a term by identifying a genus term and one or more difference words that when combined convey the meaning of the term being defined.
- "genus" simply means a relatively larger class and "species" means a relatively smaller subclass of the genus.
- The specific difference or "difference" is the attribute or attributes that distinguish the various species within a genus.

Examples

1. *"Husband" means married man.*
2. *"Skyscraper" means very tall building.*
3. *"Tent" means a collapsible shelter made of canvas or other material that is stretched and sustained by poles.*

3.5. CRITERIA FOR LEXICAL DEFINITIONS

1. **Should be grammatically correct!**
2. **Should mention the essential meaning of a term!**
3. **Should be neither too broad nor too narrow!**
4. **Should avoid circularity!**
5. **Should be affirmative not negative!**
6. **Should avoid figurative, obscure, vague and ambiguous language!**
7. **Should avoid affective terminology!**
8. **Should mention the pertinent context that the term is applied to!**